Module 5 :: Design of Combinational Circuits

<u>Practice Problems</u>

1. Design an 8-to-1 multiplexer using two 4-to-1 multiplexer and one 2-to-1 multiplexer

2. Draw the truth table of a 4-to-1 multiplexer, and show its realization using NAND gates only.

3. Realize the following functions using an 8-to-1 multiplexer, and additional gates if required:
   a) $F(A,B,C) = A'BC + AB'C + ABC' + ABC$ [sum output of full adder]
   b) $F(A,B,C) = AB + BC + CA$ [carry output of full adder]
   c) $F(A,B,C,D) = ABC' + BCD + A'D'$
   d) $F(A,B,C,D) = \Sigma\,(0, 2, 5, 7, 8, 11, 14, 15)$
   e) $F(A,B,C,D) = \Sigma\,(1, 2, 4, 7, 9, 13, 15) + \Sigma_{\Phi}\,(0, 6, 11, 14)$

2. Design a 2-bit multiplier that takes two 2-bit numbers as inputs, and produces a 4-bit product as output, using NAND gates.

4. Design a 7-to-128 decoder using the X-Y decoding technique, and estimate the number of transistors required for the implementation.

5. Suppose that a full adder is implemented using AND, OR and NOT gates only, with the AND-OR circuits in 2-level realization. The delay of an n-input AND gate is 5n nanoseconds, delay of an n-input OR gate is (5n+ 2) nanoseconds, and the delay of a NOT gate is 3 nanoseconds. What will be the worst-case propagation delay to generate the sum and carry outputs? Show the steps of your calculation.

6. What will be the worst-case propagation delay of a 16-bit ripple-carry adder using full adders as specified in Q5.

7. Repeat Q6 for a 4-bit carry lookahead adder.

8. Design a 4-to-16 decoder using 2-to-4 decoders as basic building blocks. Using the data provided in Q5, estimate the delay of the decoder circuit.